GUJARAT ENERGY DEVELOPMENT AGENCY (GEDA) · HACKATHON 2026

Mission LiFE Gujarat & EcoVerse

Production-Grade Interactive Citizen Engagement & Clean Energy Platform

Problem ID: HC26-GEDA01 · Theme: Green Energy & Sustainability · Version: 1.0.0

EXECUTIVE SUMMARY & CORE MISSION

LiFE Gujarat operationalises Prime Minister Narendra Modi's Mission LiFE (Lifestyle for Environment) across the state of Gujarat. The platform bridges the divide between macro-level climate pledges and micro-level daily habits through four flagship modules and the EcoVerse gamified companion. Built with strict TypeScript, pure unit-tested mathematical engines, real GERC tariffs, Esri satellite geometry, Open-Meteo irradiance, and zero fake appliance sliders.

1. Problem Context & Architectural Principles

[Section 1]

While general awareness of climate change is high among Gujarat citizens, everyday consumers face severe friction in adopting sustainable habits: (1) Awareness-to-Action Gap without personalised advice, (2) One-Size-Fits-All Fallacy ignoring the constraints of students vs renters vs homeowners, (3) Invisible Impact where conservation efforts feel unrewarded, and (4) Habit Fatigue without engaging social feedback loops.

The Three Non-Negotiable Engineering Rules:

- **Zero Manual Sliders:** Every calculation derives strictly from official DISCOM bills, client-side OCR, GPS telemetry, or live physical APIs.
- **Pure Mathematical Engine:** All calculations live in decoupled, unit-tested engine modules (/lib/engine/*) with 100% test coverage.
- **Traceable Regulatory Data:** Tariffs derive directly from GERC 2024–25 orders; emission factors use CEA Version 20 (0.82 kg CO2/kWh); solar subsidies follow Central PM Surya Ghar CFA guidelines (up to Rs. 78,000 cap).

2. System Architecture & Tech Stack

[Section 2]

The platform is engineered as a mobile-first Next.js 14 App Router application with dual SQLite/PostgreSQL support via Prisma ORM, strict TypeScript, Tailwind CSS, Leaflet GIS mapping, Turf.js geospatial analysis, Tesseract OCR, and full Vitest unit testing suites.

- **Frontend & State:** Next.js 14 App Router, React 18, Tailwind CSS, Lucide Icons, Recharts, jsPDF.
- **Pure Engine Layer:** /lib/engine (carbon.ts, tariff.ts, subsidy.ts, solar.ts, mobility.ts, nudges.ts, credits.ts).
- **Geospatial & OCR:** Esri World Imagery, Turf.js spatial geometry, client-side Tesseract.js worker with zero PII retention.
- **State Database:** Prisma ORM with seeded municipal wards across Ahmedabad, Gandhinagar, Surat, and Vadodara.

3. Module A: 30-Second Carbon Audit & GERC Tariff Engine [Module A]

Module A enables citizens to determine their household baseline footprint in under 30 seconds, comparing their emissions against official Gujarat benchmarks (Gujarat average: 165 kg CO2/capita/mo; Mission LiFE target: 132 kg CO2/capita/mo).

Key Technical Capabilities:

- **Auto-Resolving PIN Code:** Entering a 6-digit Gujarat PIN code automatically maps the municipal corporation, ward ID, and licensed DISCOM (Torrent Power, UGVCL, DGVCL, MGVCL, PGVCL).
- **Client-Side OCR Bill Scanner:** Users upload PDF or JPG electricity bills. In-browser Tesseract.js OCR parses consumer number, billing cycle kWh, and DISCOM with regex pattern matching. Zero consumer data leaves the browser.
- **GERC Slab-Jump Alert Engine:** Applies official 2024–25 GERC residential slab rates. Detects when a household is within 10% of jumping to a higher slab (e.g. 100 units or 250 units) and projects exact rupee savings for shaving load.
- **Multi-Factor Carbon Ledger:** Calculates composite footprint including cooking fuel (PNG piped gas vs LPG cylinders) and commute distance.

4. Module B: 'Roof-to-Rupee' Solar Polygon Studio [Module B]

Module B converts passive interest in rooftop solar into concrete action under the PM Surya Ghar: Muft Bijli Yojana using high-resolution satellite imagery.

Key Technical Capabilities:

- **Satellite Polygon Tracing:** Interactive Leaflet map utilizing Esri World Imagery (0.3m resolution). Citizens click vertices to trace their rooftop perimeter with real-time area calculation.
- **Turf.js Geometry Engine:** Enforces non-self-intersecting polygons, calculates true geodesics, and applies GEDA's 70% usable rooftop solar area heuristic.
- **Live Open-Meteo Irradiance:** Fetches live Global Horizontal Irradiance (GHI) for the exact latitude and longitude, accounting for seasonal sun angles.
- **PM Surya Ghar Subsidy Engine:** Computes Central Financial Assistance (CFA): Rs. 33,000 for 1 kW, Rs. 66,000 for 2 kW, and capped at Rs. 78,000 for systems >= 3 kW.
- **25-Year Cash Flow & Break-Even:** Models 0.5% annual PV degradation, 3% tariff escalation, inverter replacement at Year 10, and generates an interactive Recharts break-even visualization.
- **Empanelled Vendor Directory & Lead Generation:** Connects citizens with 12 GEDA-registered solar vendors across Gujarat with direct quotation requests and instant PDF feasibility reports.

5. Module C: Live GPS Mobility Tracker & Weather Nudges [Module C]

Module C verifies sustainable citizen travel (walking, cycling, electric bus, metro) using device GPS telemetry, filtering noisy readings and calculating real-time carbon offsets.

Key Technical Capabilities:

- **Live Telemetry & Haversine Smoothing:** Captures continuous GPS coordinates with 3-point rolling median speed filtering to eliminate jitter and GPS multipath drift.
- **Automated Mode Classification:** Classifies travel into WALKING (<= 7 km/h), CYCLING (7–22 km/h), BUS_TRANSIT (22–45 km/h), or TRAIN/METRO (> 45 km/h).
- **Desk-Judging 5x Commute Simulation:** Features deterministic route replay (Ahmebabad BRTS Janmarg corridor from Vastrapur to RTO) for seamless judging without requiring outdoor walking.
- **Contextual Weather Nudges:** Listens to Open-Meteo temperature, humidity, and solar conditions. If ambient temperature < 28C, nudges turning off AC; during peak solar (11 AM–2 PM), triggers shifting heavy appliance use.

6. Module D: State Climate Wallet & Command War-Room [Module D]

Module D provides the closing habit loop for citizens and high-altitude administrative control for GEDA state directors.

Citizen Climate Wallet:

- **Append-Only Credit Ledger:** Every audit (+50), solar design (+30), and verified green commute (10 credits/kg CO2) records immutable ledger entries.
- **Daily Habit Streaks & Freezes:** Features 7-day calendar heat-strips, milestone bonus multipliers at 7/30/60 days, and streak freezes for travel.
- **Proof-of-Habit Deduplication:** 64-bit dHash perceptual image hashing ensures citizens cannot upload duplicate photos to game credit awards.
- **Seeded Rewards & Official Certificate:** Redeem credits for Ahmedabad BRTS transit passes, solar discounts, or download an official GEDA Commendation Certificate PDF.

GEDA State Command War-Room (/war-room):

- **42 Municipal Ward Scorecards:** Real-time composite LiFE rankings across Ahmedabad, Surat, Vadodara, and Gandhinagar.
- **Solar Conversion Funnel:** Tracks pipeline from rooftop polygon drawn -> lead submitted -> empanelled vendor quotation.
- **Statewide Telemetry & CSV Export:** GEDA administrators can filter by city and export clean CSV datasets for policy planning.

7. Feature Innovation: The EcoVerse Companion

[EcoVerse]

EcoVerse is the personal sustainability companion proposed in the hackathon concept document. It introduces persona-adapted daily missions, fair community circles, and a dynamic living virtual green world.

Core EcoVerse Innovations:

- **Living Situation Personas:** Tailors missions according to whether the user is a Hostel Student (device control only), Rented Apartment Resident (appliance control), or Independent Homeowner (full rooftop & grid control).
- **EcoTwin Living SVG Island:** An animated virtual Gujarat ecosystem that visibly evolves across 5 tiers: Level 1 (Barren Atoll) -> Level 2 (Sprouting Haven) -> Level 3 (Solar Oasis) -> Level 4 (Renewable Haven with animated spinning wind turbine & electric BRTS) -> Level 5 (Utopian Gujarat EcoVerse). Clickable landmarks reveal real-time impact.
- **Daily EcoQuests with Swap & Skip:** 3 daily targeted tasks with context rationales. Citizens can complete, skip without penalty, or swap tasks if unsuitable for their day.
- **Community Circles & Invite Codes:** Enables joining groups via codes like IITGN-HOSTEL-4 or VASTRAPUR-GREENS. Employs a fair-play 'Most Improved' track so students and flat residents have equal opportunity to top the leaderboard.
- **Tangible Impact Storytelling:** Translates abstract emissions into understandable metrics: Neem trees planted, educational classroom ceiling fan hours powered, and thermal coal displaced.

8. Verification & Quality Benchmarks

[Audit Trail]

The application adheres to the highest engineering standards as validated by automated test runners:

- **TypeScript Strict Mode:** 100% type safety with zero 'any' escapes in core calculation libraries. Ran 'tsc --noEmit' with 0 errors.
- **Unit Test Coverage:** Vitest suite covering carbon, tariffs, solar cash flow, mobility smoothing, and credit ledgers. 48/48 tests passing (100% pass rate).
- **Next.js Production Build:** All 28 static and dynamic routes pre-rendered and verified via 'next build'.
- **Offline & Demo Robustness:** Every API features graceful fallbacks with visible badges; includes synthetic commute traces and pre-packaged DISCOM bills for offline hackathon demonstration.

DELIVERED FOR GUJARAT ENERGY DEVELOPMENT AGENCY (GEDA)

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